ZADU.js: Measuring and Visualizing distortions in dimensionally reduced data

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Problem

Dimensionality reduction (DR) techniques like UMAP project high-dimensional data into 2D for visualization — but every projection inevitably distorts the original geometry. How can users discover, quantify, and reduce these distortions interactively in the browser?

Idea

Using the existing python library known as zadu, reimplementing those metrics in JavaScript to allow us to compute them on the browser. Specifically the trustworthiness and continuity metrics which measure false (Not meant to be a neighbor but it) and missing (Meant to be a neighbor but isn't) neighbors respectively.

State of the Art

CheckViz is a visualizer for these distortion measures which has been implemented in python with the use of the matplotlib library and server side python. Other approaches include - Warping Index (by Jaume Ros) and Hypertrix (by Shivam Raval and Fernanda Viegas)

Where to find these metrics?

npm: https://www.npmjs.com/package/zadu-js
Github: https://github.com/jonathantarun/zadu-js

DEMO:

Check out this demo showcasing these metrics on a sample dataset of 200 points from MNIST.

SCAN ME

Distortion Types

False Neighbors: Points appear close in the 2D embedding but were far apart in the original high-dimensional space.

Missing Neighbors: Points that were genuinely nearby in the original space have been pulled apart in the embedding.

Phantom Clusters: Groups that appear as distinct clusters in the 2D embedding but were not actually separated in the original high-dimensional space.

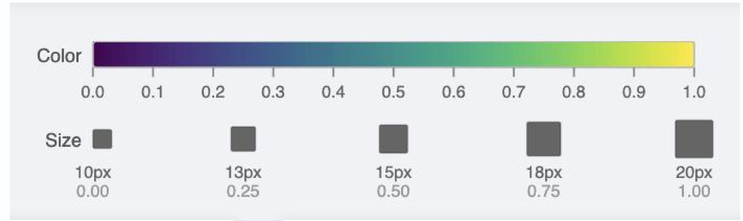
Dispersed Clusters: Points that belonged to the same real cluster in the original space have been scattered across different regions in the embedding.

Point-level

Metric	Measures	Distortion
Trustworthiness (T)	Are embedding neighbors real neighbors?	False neighbors
Continuity (C)	Do real neighbors stay close in embedding?	Missing neighbors

Cluster-level (HDBSCAN + SNN distance)

Metric	Measures	Distortion
Steadiness (S)	Are embedding clusters real clusters?	Phantom clusters
Cohesiveness (Co)	Do real clusters stay together in embedding?	Dispersed real clusters



Per-point quality map of a 200-sample MNIST embedding (UMAP, n_neighbors=15, min_dist=0.10). Color and size encode per-point Trustworthiness (Viridis color scheme). Point 70 (digit 7) has the lowest T score (0.505) indicating that it has many false neighbors. Minimum scores — T=0.505, C=0.440, S=0.228, Co=0.086 — show that cluster-level metrics reveal far more distortion than point-level scores alone.

References

1. Venna, J. & Kaski, S. (2006). Local multidimensional scaling. Neural Networks, 19(6), 889–899.
2. Jeon, H., Ko, H.-K., Kim, Y., Jo, J., & Seo, J. (2021). Measuring and Explaining the Inter-Cluster Reliability of Multidimensional Projections. IEEE TVCG.
3. Jeon, H., Cho, A., Jang, J., Lee, S., Hyun, J., Ko, H.-K., Jo, J., & Seo, J. (2023). ZADU: A Python Library for Evaluating the Reliability of Dimensionality Reduction Embeddings. IEEE VIS 2023, pp. 196–200.